

JKBOSE

2026 CLASS 11TH

MOTION IN PLANE

LECTURE-3

PHYSICS

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Topics To Be Covered

- 1 Revision ✓
- 2 Uniform Circular Motion ✓
- 3 Acceleration in uniform circular motion ✓ (N/m)
- 4 Subtraction of two vector ✓
- 5 .

"Small consistent actions shape
a powerful future—far more
than rare bursts of motivation
ever will."

Recap from 1st Lecture

- Motion in 1D



- Motion in 2D



- Motion in 3D



- Displacement in 2D

$$\vec{r} = \vec{x} + \vec{y}$$

- Velocity in 2D

$$\vec{v} = \vec{v}_x + \vec{v}_y$$

- Acceleration in 2D

$$\vec{a} = \vec{a}_x + \vec{a}_y$$

- Equation of motion

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

$$v_x = u_x + a_x t$$

$$s_x = u_x t + \frac{1}{2}a_x t^2$$

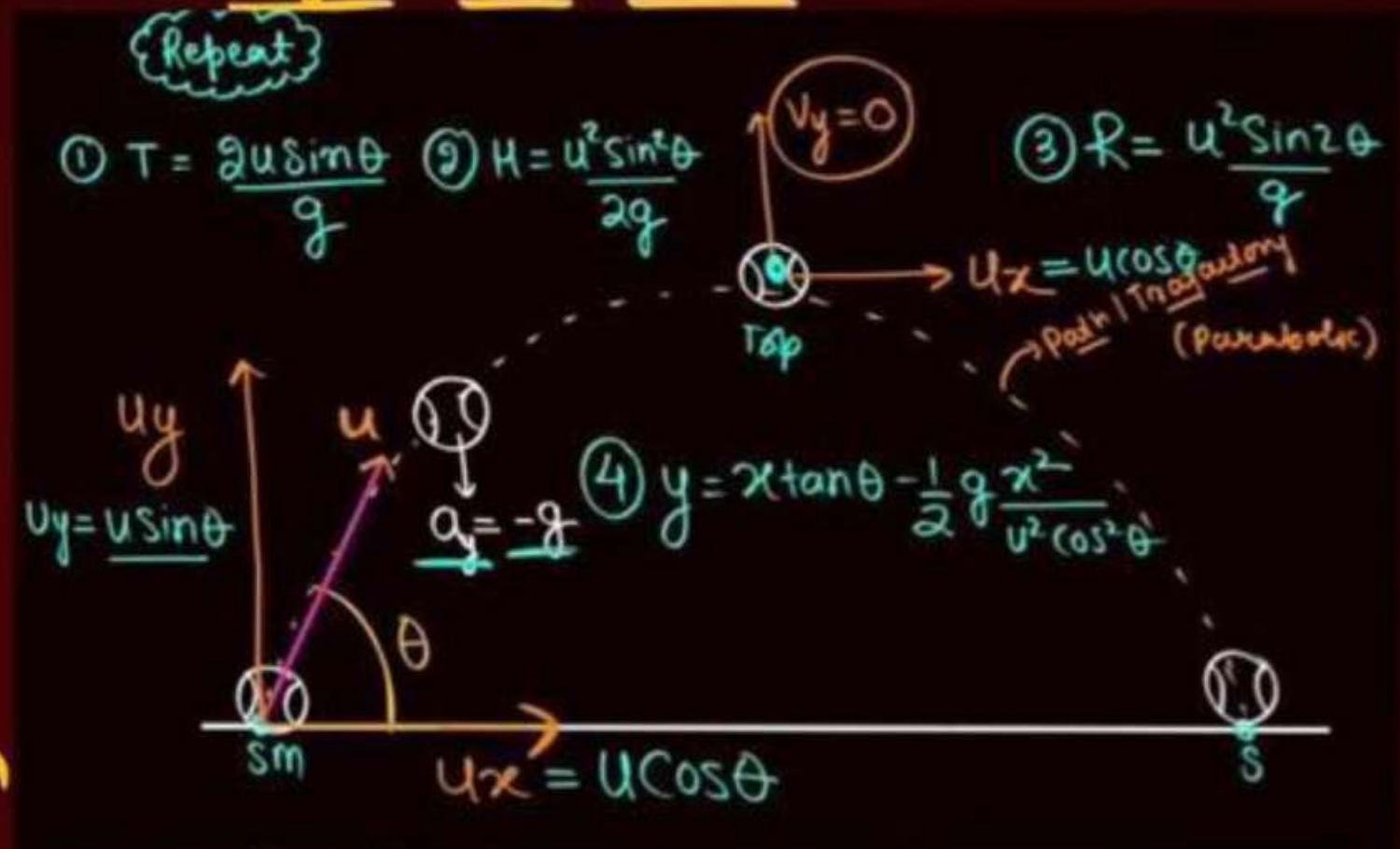
$$v_x^2 = u_x^2 + 2a_x s_x$$

$$v_y = u_y + a_y t$$

$$s_y = u_y t + \frac{1}{2}a_y t^2$$

$$v_y^2 = u_y^2 + 2a_y s_y$$

Projectile Motion



Projectile Motion

(5-mark) 2026, 2025, 2024, 2023, 2022



- Example of 2D
- To explain the 2D motion
- when an object is thrown with Same Angle under the influence of gravity

Neet level

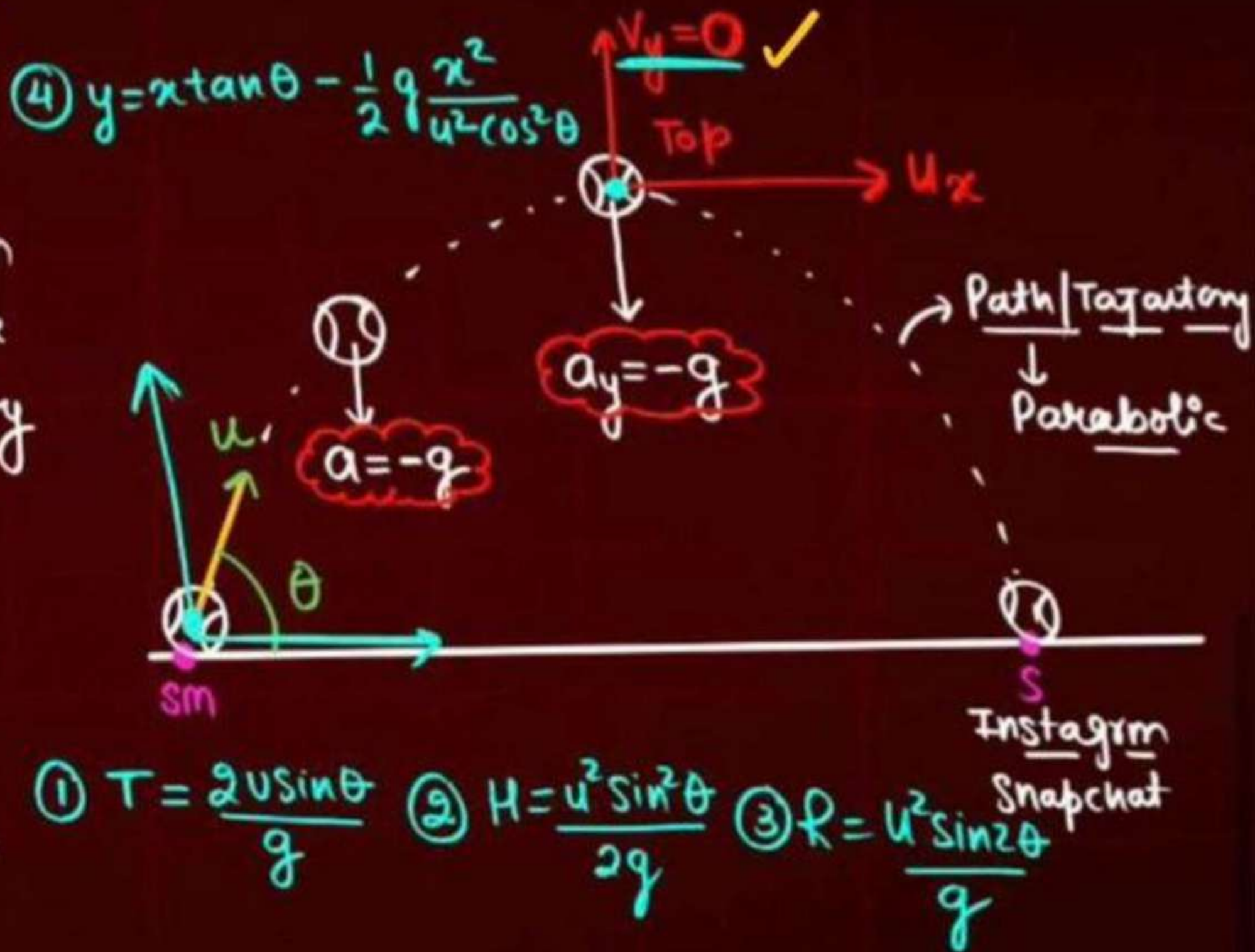
$a_y = -g = -9.8 = -10 \text{ m/s}^2$

$a_x = 0$

2024 JKBOSE

$v_y = 0$

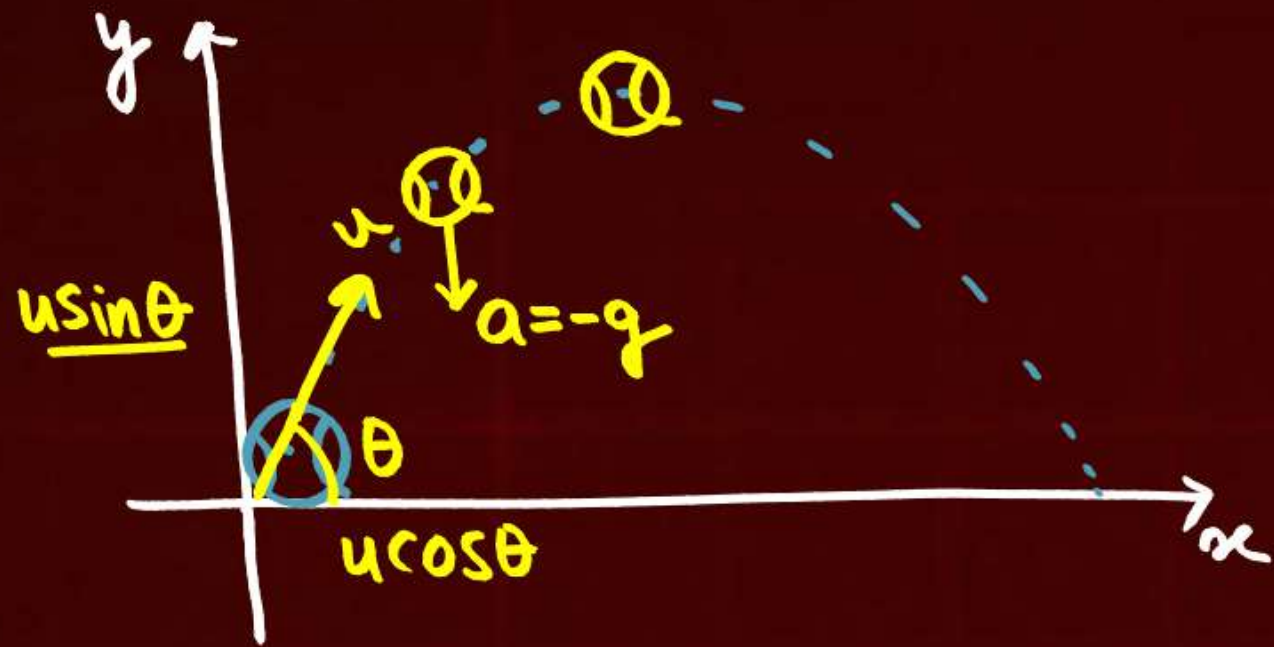
Path | Tangency → Parabolic



Recap from 2nd Lecture



Projectile Motion



① Time period.

The time for which ball remain in Air $T = \frac{2u \sin \theta}{g}$

② Maximum Height.

The maximum height attain by Projectile

$$H = \frac{u^2 \sin^2 \theta}{2g}$$

③ Range

The maximum Horizontal displacement of object $R = \frac{u^2 \sin 2\theta}{g}$

④ Equation of Trajectory $\rightarrow$ Relationship b/w y and x

$$y = x \tan \theta - \frac{1}{2} g \frac{x^2}{u^2 \cos^2 \theta}$$

Uniform Circular Motion

- 9th class (Sirf Naam ka uniform motion hai)

Circular motion

gool gool gumna

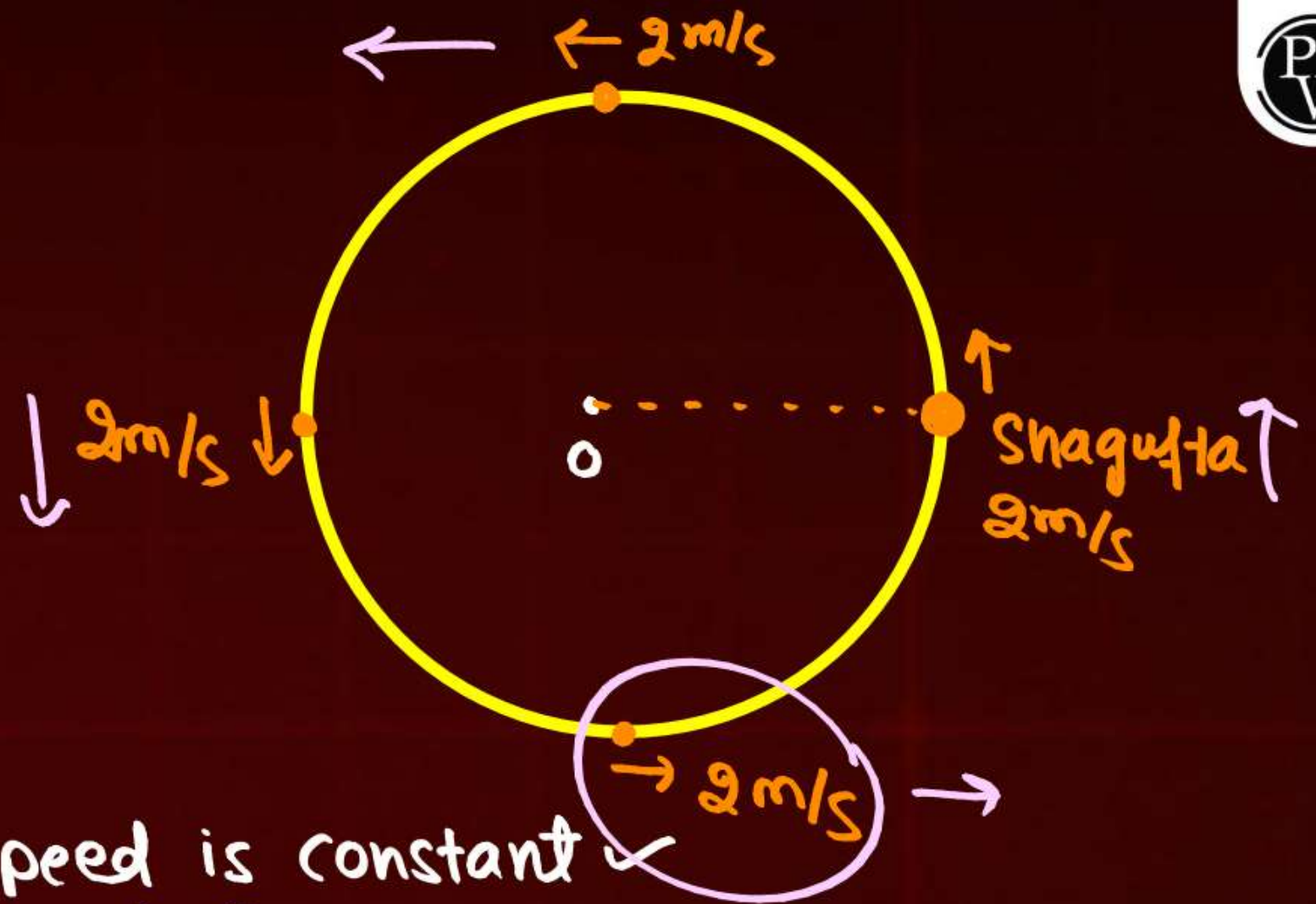
Circular motion

when a body moves in a circular path

around the fixed centre (O)

uniform circular motion

when body moves in circular path with constant speed.



① Speed is constant ✓

Speed = 2m/s
(magnitude)

② But velocity change. ✓

✓ Velocity = magnitude + Direction.
2m/s constant change

③ Non-uniform motion ✓

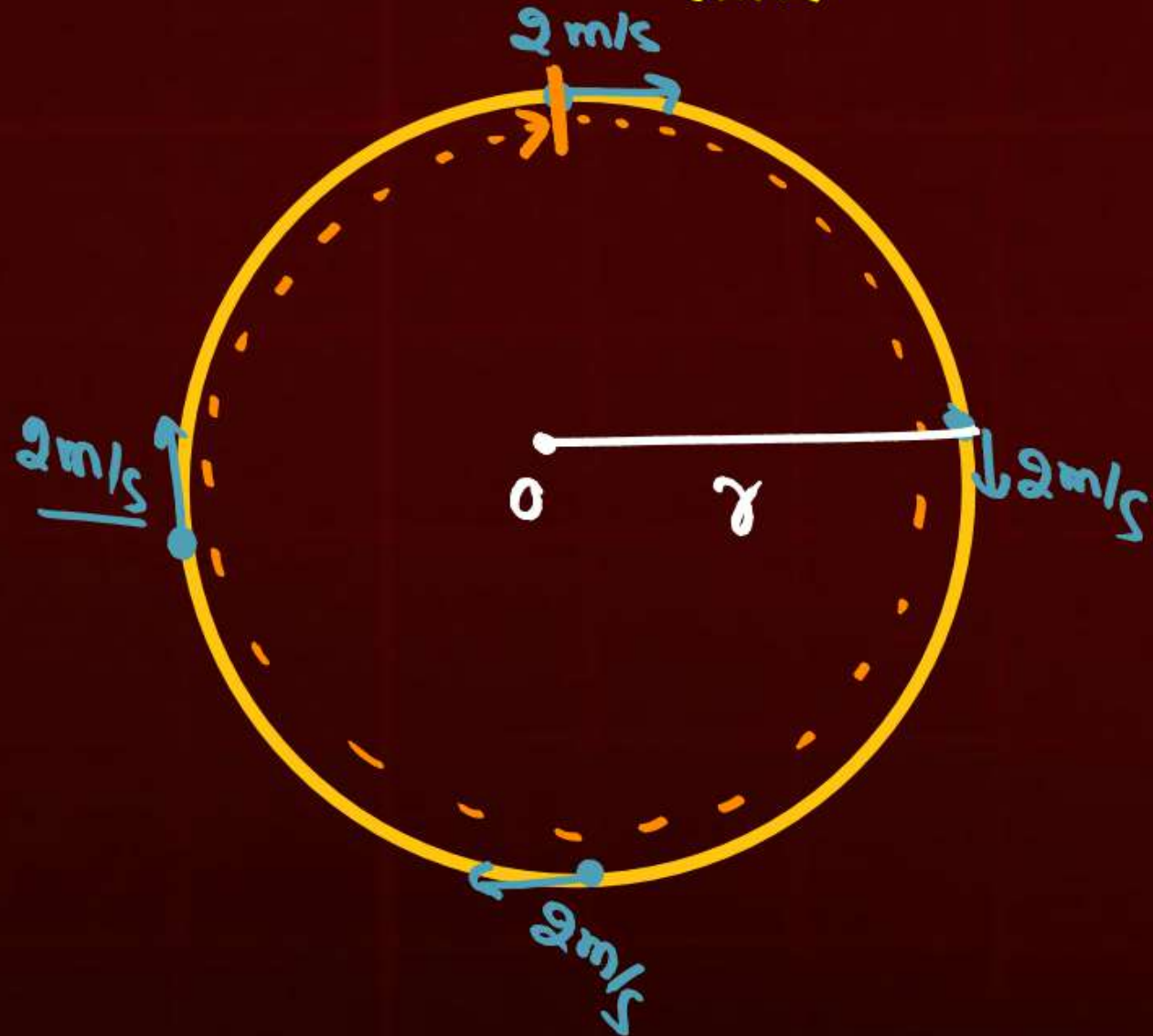
Q uniform circular motion? JKBOSE — 2016

- (a) uniform motion ✗
- (b) non-uniform motion ✓
- (c) 1D motion ✗
- (d) 3D-motion ✗

Speed in uniform Circular Motion

Speed :- Rate of change of distance

$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$



$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

$$\text{Speed} = \frac{\text{Circumference of circle}}{\text{Time}}$$

$$\underline{\underline{\text{Speed}}} = \frac{2\pi r}{T}$$

$$\text{Speed} = 2\pi r \times \left(\frac{1}{T} \right)$$

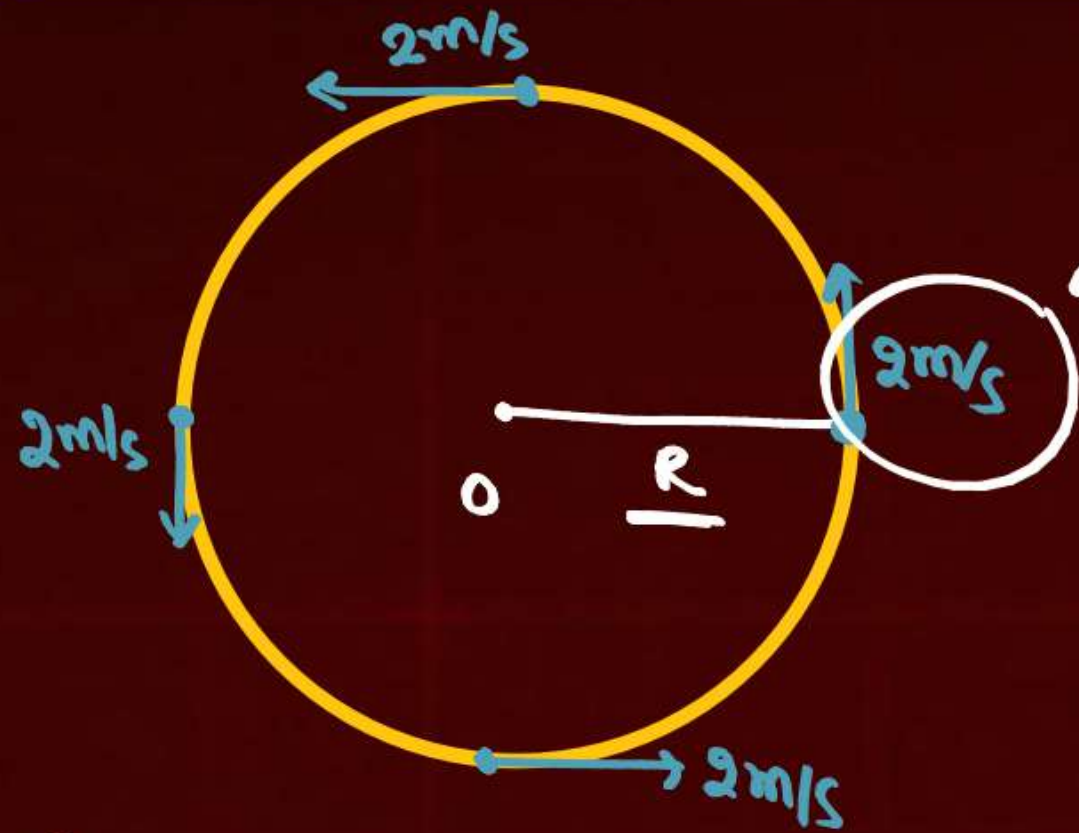
$$\text{Speed} = 2\pi r \nu$$

$$\frac{1}{T} = \nu \downarrow \text{frequency}$$

(13) chapter

Acceleration in uniform circular motion

→ Centripetal acceleration.



“Jaha velocity change hogi waha acceleration hoga”

Centripetal Acceleration

The acceleration which changes only direction of velocity.

→ Derivation (NLM) ✓
5 marks ✓

① Speed = $\underline{2\text{m/s}}$ → constant.
(magnitude)

② Velocity = $\underline{2\text{m/s}}$ + Direction
↓
change magnitude

③ Non-uniform (motion)
↳ acc. Hota hai

$$a_c = \frac{v^2}{R}$$

$$F = ma$$

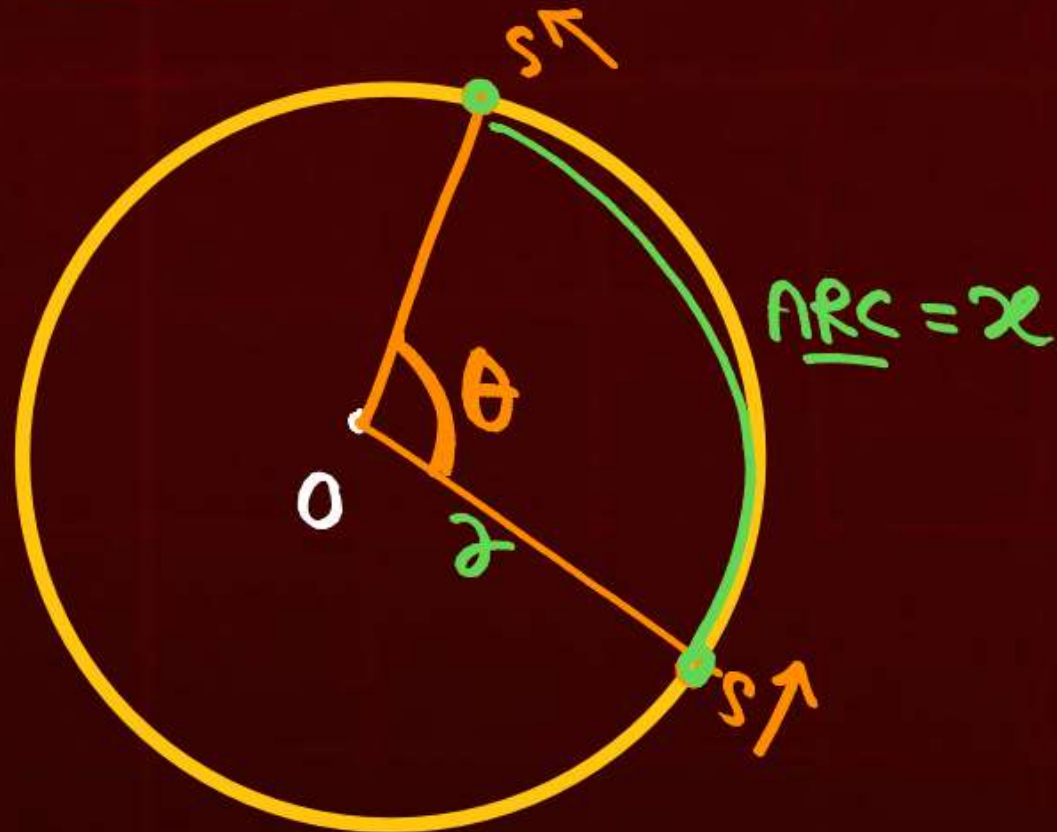
$$F_c = ma_c$$

$$F_c = m \frac{v^2}{R}$$

Acceleration in uniform circular motion

1. Angular Displacement (θ) ^{→ Rotation}

The Angle Rotated in Circular motion



$$\# \underline{\text{Angle}} = \frac{\text{ARC}}{R}$$

$$\theta = \frac{x}{r}$$

$$\boxed{r\theta = x}$$

Relation ship b/w θ and $\underline{x}$
 $\downarrow$
 Angular displacement
 $\uparrow$
 Linear displacement

Acceleration in uniform circular motion

2. Angular Velocity (ω) ^{omega} 2016

The Rate of change of Angular displacement (θ)

$$\omega = \frac{\theta}{t} \quad \theta = \frac{x}{r}$$

$$\omega = \frac{\left(\frac{x}{r}\right)}{t}$$

$$\omega = \frac{x}{rt}$$

$$r\omega = \left(\frac{x}{t}\right)^{\vee}$$

$$r\omega = v$$

$$v = r\omega$$

linear velocity ✓ angular velocity ✓

$$v = r\omega$$

$$\boxed{\frac{v}{r} = \omega}$$

*

Acceleration in uniform circular motion

3. Angular Acceleration (α) ^{→ Alpha} 2017

Rate of change of Angular
Velocity

$$\alpha = \frac{\omega}{t} \quad \omega = \frac{v}{r}$$

$$\alpha = \frac{v}{r t}$$

$$r \underline{\alpha} = \left(\frac{v}{t} \right)^a$$

$$r \alpha = a$$

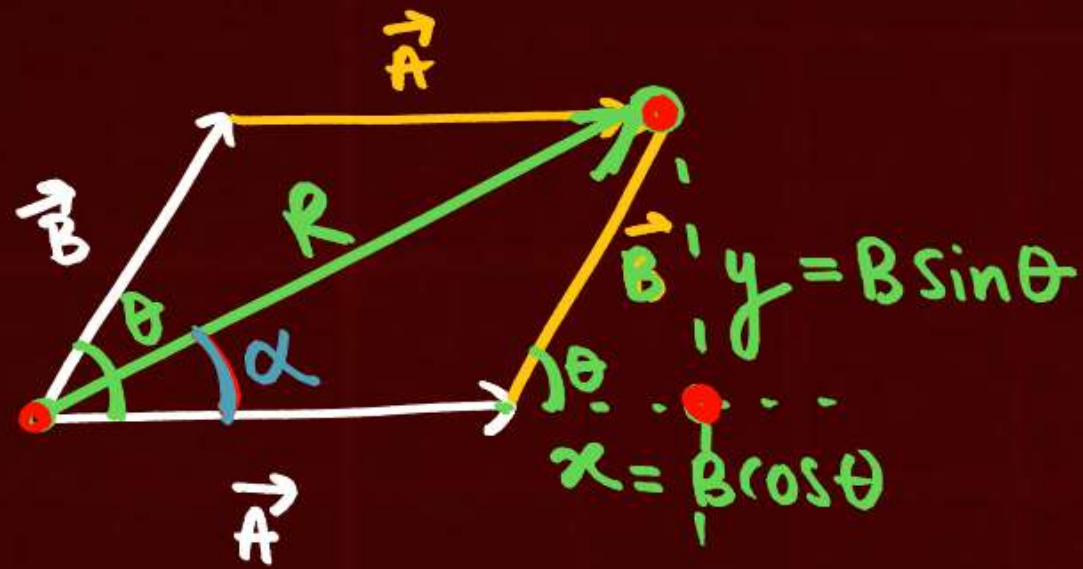
$$a = r \alpha \rightarrow \text{angular acceleration} \underline{\underline{=}}$$

acceleration ✓

Subtraction of two vector

Addition of two vector

$$\vec{R} = \vec{A} \oplus \vec{B}$$

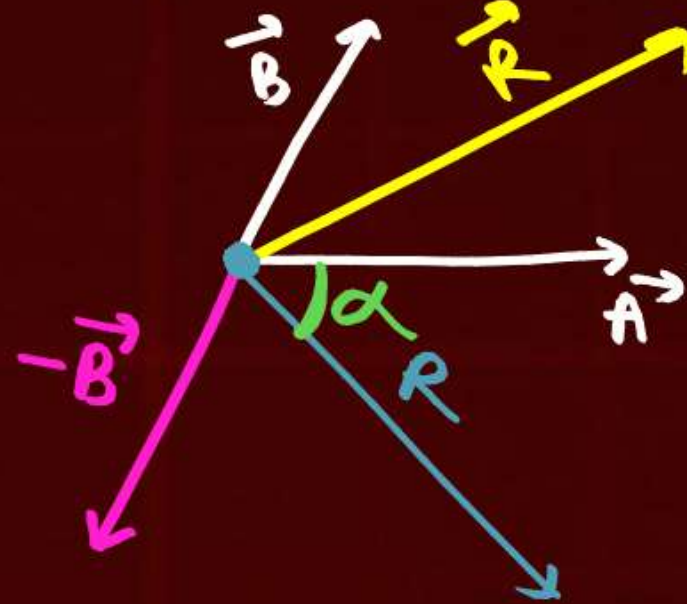


$$R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

$$\tan \alpha = \frac{B \sin \theta}{A + B \cos \theta}$$

Subtraction of two vector

$$\vec{R} = \vec{A} \ominus \vec{B}$$



$$R = \sqrt{A^2 + B^2 - 2AB \cos \theta}$$

$$\tan \alpha = \frac{B \sin \theta}{A - B \cos \theta}$$

SUMMARY

uniform circular motion

Speed

$$\# a_c = \frac{v^2}{r}$$

$$\# r\dot{\theta} = v$$

$$r\dot{\omega} = v$$

$$r\dot{\alpha} = a$$

Subtraction of two vector

HOMework

Copy - check ✓ → cp ✓

Get Ready for Gabber numerical series ✓ (Live) ✓

Thank
You